UNIT 1 • MINI PROJECT

Enterprise Student Record and Academic Management Platform

Design Thinking • Product Planning • MVP • Roadmap

01	02	03
Understand	Design	Plan

Prepared as a Unit-1 product-engineering mini project

1. Project Overview

The Enterprise Student Record and Academic Management Platform is a planned software solution for managing student records and academic information within a higher education institution. The project follows the Unit-1 focus on business understanding, Design Thinking, stakeholder analysis, requirements engineering, product vision, MVP definition, roadmap, project planning and initial architecture.

Primary users	Core problem	MVP focus	Approach
Students, Faculty, Administrators, Exam Cell, IT	Fragmented/manual academic records	Registration, records, attendance, reports	Design Thinking + Product Engineering

2. Objective

- Study the academic management process before beginning development.
- Identify operational problems in student admissions, course registration, attendance, examination and result management.
- Apply Empathize, Define, Ideate, Prototype and Test to understand and shape the solution.
- Translate stakeholder needs into functional and non-functional requirements.
- Define a practical first release (MVP) and a phased product roadmap.

3. Business Understanding

The academic management domain includes student admissions and registration, course registration, attendance tracking, examination processing, result management and academic report generation. A centralized platform is proposed to reduce dependency on fragmented manual records and improve visibility for students, faculty and academic administrators.

3.1 Key Business Areas

Business Area	Purpose
Student Registration	Maintain student identity, academic and registration records.
Course Management	Manage course registration and academic information.
Attendance	Record and monitor student attendance.
Examination&Results	Organize examination information and result records.
Reporting	Generate academic reports for relevant stakeholders.

4. Problem Analysis

The project starts from the problems described in the assignment: manual record management can make academic information difficult to organize, update and report. The platform therefore focuses on centralized academic information and clearer workflows rather than starting with code.

4.1 Problem → Impact → Product Response

Problem	Impact	Proposed response
Fragmented/manual student records	Difficulty maintaining consistent academic information	Centralized student record management
Manual attendance handling	Extra effort in tracking and reviewing attendance	Attendance management module
Scattered examination/result information	Slower access to academic status and reports	Result and academic record module

Manual report preparation	Time-consuming academic reporting	Academic report generation
Different stakeholder needs	Poor visibility and unclear responsibilities	Role-oriented platform access

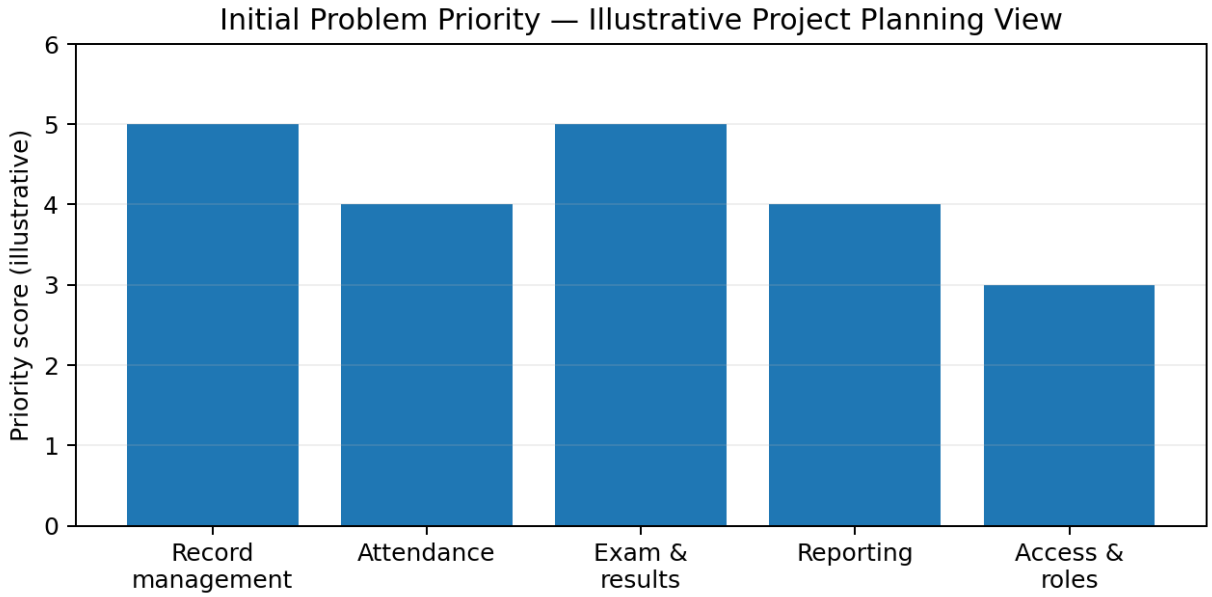


Figure 1. Illustrative prioritization for planning; not a measured institutional survey.

5. Design Thinking

The project applies the five-stage Design Thinking methodology specified in the assignment. The emphasis is on understanding users and validating the product direction before detailed development.

Stage	Application to this project
1. Empathize	Understand students, faculty and administrators, their workflows, goals and pain points.
2. Define	Convert observations into a clear academic-management problem statement.
3. Ideate	Generate solution ideas for records, attendance, results and reporting.
4. Prototype	Create low-fidelity screens/workflows for key modules.
5. Test	Validate navigation, clarity and task completion with intended users.

5.1 Core Problem Statement

“Students, faculty and academic administrators need a centralized and reliable way to manage student records and academic information because fragmented or manual processes can make registration, attendance, examination, result management and reporting harder to track and access.”

6. Stakeholder Analysis&User Personas

Stakeholder	Role	Key expectation
Students	Access records, attendance, results and academic information.	Easy access, clarity and timely information.
Faculty Members	Maintain academic and attendance information.	Efficient updates and accurate student data.

Academic Administrators	Manage institutional academic records and reports.	Centralized visibility and reliable reporting.
Examination Cell	Handle examination and result-related information.	Accurate and organized result records.
IT Support	Support platform availability and technical operations.	Security, reliability and maintainability.

6.1 User Personas

Persona	Goal	Pain point	Platform support
Aarav — Student	View academic information quickly; track attendance and results.	Unclear or delayed access to academic records.	Student dashboard, record view, attendance and reports.
Prof. Mehta — Faculty	Update attendance and academic information efficiently.	Repeated/manual record handling.	Faculty dashboard and attendance/academic update tools.
Dr. Sharma — Administrator	Monitor academic records and generate reports.	Scattered information and manual reporting effort.	Administrative dashboard and reporting module.

7. Requirements Engineering

7.1 Functional Requirements (FRS)

ID	Requirement	Description
FR-01	Student Registration	The platform shall maintain student registration and profile information.
FR-02	Student Record Management	The platform shall store and manage academic records.
FR-03	Course Registration	The platform shall support course registration information.
FR-04	Attendance Management	The platform shall allow attendance information to be recorded and reviewed.
FR-05	Examination&Results	The platform shall manage examination and result information.
FR-06	Academic Reporting	The platform shall generate academic reports from managed information.

7.2 Non-Functional Requirements (NFRS)

ID	Quality attribute	Requirement
NFR-01	Security	Protect student and academic information through appropriate authentication and access control.
NFR-02	Usability	Provide a clear and easy-to-use interface for students, faculty and administrators.
NFR-03	Reliability	Maintain dependable access to academic information and minimize data loss.
NFR-04	Scalability	Support growth in users, records and academic information.

8. IBM Design Thinking

1. Focus on User Outcomes — measure success by useful academic outcomes for students, faculty and administrators rather than by technology alone.
2. Restless Reinvention — treat the product as continuously improvable using feedback and observed usage.
3. Diverse Empowered Teams — bring together academic stakeholders, product/design roles, developers, QA and IT support.

8.1 IBM Loop: Observe → Reflect → Make

OBSERVE	REFLECT	MAKE
Study user interactions, academic workflows and operational friction.	Analyze feedback, identify priorities and re-evaluate requirements.	Design improvements, implement changes and prepare the next iteration.

9. Product Vision

Vision Statement: To provide a centralized, user-focused and reliable platform that simplifies student record and academic information management for higher education stakeholders.

9.1 Business Objectives

- Centralize student and academic information.
- Simplify student, faculty and administrative workflows.
- Improve accessibility of attendance and result information.
- Support organized academic reporting.
- Create a scalable foundation for later development and DevOps activities.

9.2 Success Criteria

Area	Success criterion
Usability	Users can understand and navigate the core workflows.
Completeness	MVP covers the required first-release academic functions.
Data organization	Student and academic information is managed in a structured way.
Reporting	Core academic information can be converted into useful reports.
Future readiness	The product structure can support later Agile and engineering phases.

10. Minimum Viable Product (MVP)

The assignment specifies the first release around student registration, record management, attendance and academic report generation. Examination/result information and course-related information are included in the broader academic platform requirements.

ID	Module	Scope	Priority
MVP-01	Student Registration	Create and manage student registration/profile information.	**Must have**
MVP-02	Record Management	Store and manage essential academic records.	**Must have**
MVP-03	Attendance	Record and review student attendance.	**Must have**

MVP-04	Academic Reports	Generate basic academic reports.	**Must have**
MVP-05	Result Information	Support basic examination/result information within the academic record flow.	**Core extension**

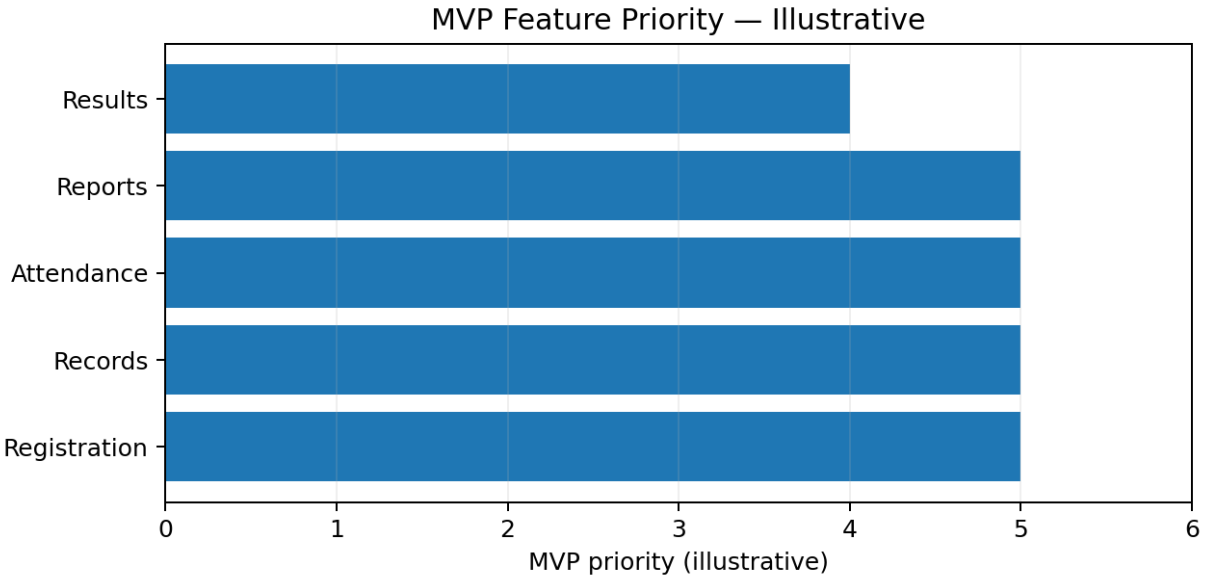
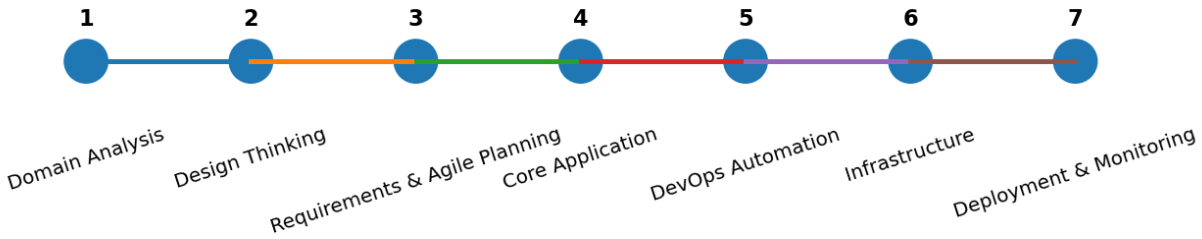


Figure 2. Illustrative MVP prioritization based on the assignment's required first-release scope.

11. Product Roadmap

Phase	Focus	Outcome
Phase 1	Domain Analysis	Understand academic processes, stakeholders and pain points.
Phase 2	Design Thinking	Empathize, define, ideate, prototype and test.
Phase 3	Requirements&Agile Planning	Finalize FRS/NFRS, backlog and milestones.
Phase 4	Core Application	Develop frontend/backend modules for the MVP.
Phase 5	DevOps Automation	Introduce version control, testing and CI/CD practices.
Phase 6	Infrastructure	Prepare deployment and infrastructure automation as the course progresses.
Phase 7	Deployment&Monitoring	Move toward production deployment and monitoring.

Enterprise Product Roadmap



12. Initial System Architecture Concept

The initial architecture is intentionally conceptual for Unit 1. It identifies the main layers that can support later implementation without prematurely locking the project to a specific technology stack.

Layer	Responsibilities
User Layer	Student • Faculty • Administrator • Examination Cell
Application Layer	Authentication • Student Records • Course Registration • Attendance • Results • Reports
Data Layer	Student data • Academic records • Attendance • Examination/result data
Engineering Layer	Version control • Testing • CI/CD • Deployment • Monitoring

13. Project Repository Structure

```
enterprise-student-record-platform/
├── docs/
│   ├── BRD/
│   ├── product-vision/
│   ├── personas/
│   ├── requirements/
│   ├── design-thinking/
│   └── roadmap/
├── app/
│   ├── frontend/
│   ├── backend/
│   └── modules/
│       ├── students/
│       ├── courses/
│       ├── attendance/
│       ├── examinations/
│       └── reports/
├── database/
├── tests/
├── deployment/
└── README.md
```

14. Project Charter&Milestones

Charter item	Definition
Project Goal	Plan a student academic-management product using Design Thinking and product-engineering practices.
Scope	Business understanding, stakeholders, personas, requirements, vision, MVP, roadmap, architecture and repository planning.
Out of Scope for Unit 1	Full production implementation and advanced features not required by the assignment.
Primary Deliverables	BRD, Product Vision, Stakeholder Analysis, Personas, FRS, NFRS, Design Thinking, IBM Design Thinking, IBM Loop, MVP, Roadmap, Architecture, Charter and repository plan.

14.1 Milestone Plan

Milestone	Activity	Completion outcome
M1	Business understanding&problem analysis	Domain and problem statement completed
M2	Design Thinking&personas	Five stages and user personas documented
M3	Requirements&product vision	FRS, NFRS and vision finalized
M4	MVP&roadmap	First release and phased roadmap defined
M5	Architecture&repository	Initial architecture and folder structure prepared

15. Unit-1 Submission Checklist

- Business Requirements Document (BRD)
- Product Vision Document
- Stakeholder Analysis Report
- User Persona Report
- Functional Requirements Specification (FRS)
- Non-Functional Requirements Specification (NFRS)
- Design Thinking Report
- IBM Design Thinking Report
- IBM Loop Documentation
- MVP Definition
- Product Roadmap
- Initial System Architecture Diagram
- Project Charter
- Project Repository Initialization
- Project Folder Structure
- Product Discovery Documentation

16. Conclusion

This mini project converts the assignment requirements into a structured product-engineering plan for an Enterprise Student Record and Academic Management Platform. The work begins with business understanding and user needs, moves through Design Thinking and requirements engineering, defines a focused MVP, and establishes a

roadmap and initial architecture for later development. The document is therefore designed to be understandable as a project report while remaining aligned with the Unit-1 scope.

Source Basis&Scope Note

This report is prepared from the two supplied Unit-1 documents. The first document was used as the reference for structure, terminology and product-engineering framing; the second document defines the specific Student Record and Academic Management Platform assignment and required deliverables. Charts marked “illustrative” are planning visuals created for readability and are not claimed as measured institutional data.